MCA

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1) Diff b/w von neumann and harvard architecture

Von Neumann Architecture :-

→ Hugh level of flexibility as memory is shared between instructions and data so that level assigned to each can fluctuate based on task
→ Speed is limited when compared to harvard due to only having "1" memory location and set of buses
→ Used in general purpose computer that will be used for many different purposes.

Harvard Architecture

→ Limited flexibility as there is only a certain amount of memory that can be used for data and a certain amount for instructions
→ Two set of memory and buses mean data can be handled more quickly which decreases execution time
→ Typical in embedded system like washing machines.

2) What is Endianess? Explain type of Endianess with example.

→ Order in which a computer stores multi byte numbers in memory. It decides whether the system writes the "big end" MSB first or "little end" into 1st memory slot.

1) Big Endian :-

MSB stored at lowest memory address :- Eg 0x12345678

0x1000 → 0x12 → MSB
0x1001 → 0x34
⋮
0x1004 → 0x78 → LSB

2) Little Endian :-

LSB stored at lowest memory address

0x1000 → 0x78 → LSB
⋮
0x1004 → 0x12 → MSB

2) Features of LPC 21xx and LPC 23xx

[ARM-7]
↓
[AHB/APB]

[Flash] [SRAM] [GPIO] [Timer] [UART] [SPI/I²C] [ADC] [PWM]

[Watchdog]
↓
[CAN] → LPC 23xx only

LPC 21xx Features

→ ARM 7 32 bit RISC processor
→ 60 MHz clock
→ Single High performance bus
→ No DMA controller
→ 40 kB SRAM
→ No ethernet
→ USB 2.0
→ No storage expansion

LPC 23xx Features

→ ARM 7 TDMI
→ 72 MHz clock
→ Dual layer AHB bus matrix
→ General purpose and dedicated
→ 96 kB SRAM (Split SRAM)
→ On chip Ethernet dedicated DMA
→ USB 2.0, Host and OTG
→ SD/MMC card Interface

3) Explain GPIO program

[GPIO port] → [GPIO Register] → [IODIR, IOSET, IOCLR] → [GPIO Pin] → [LED]

```
#include <LPC21xx.h>
int main (void) {
IODIR0 |= (1<<10);
  while (1)
  {
    IOSET0 |= (1<<10); //LED ON
     delay (1000)
    IOCLR0 &= ~(1<<10); //LED off
     delay (1000);
```

IODIR → configures port/pin as I/O
IOSET → set as logic high
IOCLR → set to low
IOPIN → current status

5) Explain LED and interfacing with ARM

→ LED connected to GPIO Pin through a resistor

→ Configure the GPIO pin as O/P using IODIR0

→ Make pin high using IOSET0

→ Make pin low using IOCLR

Program same as previous.